

or the drive doesn't support the setting fully. That is the reason for choosing only a few flags as listed above and the -u flag is a crucial one, use it only if you are confident that your chipset and kernel can handle it.

Let's test speed of the drive:

```
[root@aryan 00-anup]#
hdparm -t /dev/hda
/dev/hda:
```

Timing buffered disk reads: 64 MB in 24.30 seconds = 2.63 MB/sec

We got a speed of 2.63 MB/sec as the DMA was disabled. You may try the command again to recheck. Now, let's use the flags to change drive settings and then look for any performance difference. The ideal command option is: `#hdparm -d 1 -X 69 -c 1 -M 192 -A 1 -a 64 -m 16 /dev/hda` hit [Enter] to execute the command. The output of the command shows the change in settings.

For Example:

```
[root@aryan 00-anup]#
hdparm -d 1 -X 69 -c 1 -M 192 -A 1 -a 64 -m 16 /dev/hda
Output:
/dev/hda:
setting fs readahead to
64
```

```
setting 32-bit IO_support
flag to 1
```

```
setting multcount to 16
setting using_dma to 1
```

```
(on)
setting xfermode to 69
(UltraDMA mode5)
```

```
setting drive read-
lookahead to 1 (on)
multcount = 16 (on)
IO_support = 1 (32-bit)
using_dma = 1 (on)
readahead = 64 (on)
```

```
setting acoustic
management to 192
acoustic = 192
(128=quiet ... 254=fast)
```

We have discussed '-d' flag to enable DMA earlier, so let's look at the other flags used in the above command. When enabling DMA, the '-X' flag is used to set the drive to a particular DMA mode. Here: the numeric value 69 means 64 plus the value of DMA mode 5 is UltraDMA mode 5 ATA100, add 6 for ATA133) and '-c' flag enables/disables 32 bit I/O support (the default is 16Bit). Modern hard disks have the ability to slow down

in order to reduce their noise output, so the flag '-M' is used to change the acoustic level of a drive, the numeric value used with this flag is between 128 and 254; the silent and hence the slowest is 128. Use 0 is disable or to use the default drive setting. We prefer a value of 192.

The flag '-A 1' enables the auto read-ahead feature of the drive and '-a 64' tells the drive how far to read ahead. The '-m 16' flag is a courtesy as most modern IDE drives have this feature enabled; this feature reduces operating systems overhead on a drive's I/O. Now, let's test for any difference in speed of the drive using `#hdparm -t /dev/hda` command.

```
[root@aryan 00-anup]#
hdparm -t /dev/hda
```

Output:

```
/dev/hda:
```

Timing buffered disk reads: 64 MB in 1.67 seconds = 38.32 MB/sec

You can see from the output that the speed has gone up from 2.63 MB/sec to 38.32 MB/sec.

Changing the GRUB Splash image

GRUB (GRand Unified Boot loader) is commonly used to boot OSes by almost all Linux distributions. If you are bored seeing the same blue GRUB screen every time, then take a break—you can change the GRUB 'splash' image by editing GRUB's config file. Before we proceed to replace it with another image, let's understand the image requirements.

The image has to be 14 colours with a maximum resolution of 640 x 480, and in the xpm format. The immediate question is "how do I make such an image?" Well, some distributions like Redhat 9 have a command line utility called ImageMagick. Do `#man ImageMagick` to know more about it. Let's consider that you have an image in jpeg (*.jpg) format that you want to load as your splash screen. To change 'photo.jpg' into a splash screen image switch to root (administrator) then use

the following command in konsole or xterm `#convert -depth 8 -colors 14 -resize 640x480 /root/grub-pics/photo.jpg /root/grub-pics/photo.xpm` assuming that your image is in a folder called 'grub-pics' under /root. Needless to say that convert command converts the image in the necessary format but we still have to compress it to 'gz' format; so under `/root/grub-pics` do `#gzip photo.xpm`. The output of the command is the file we need 'photo.xpm.gz'. Now, copy the image to `/boot/grub` `#cp /root/grub-pics/photo.xpm.gz /boot/grub`

The last step is to edit GRUB's config file. Open the `grub.conf` file in `/boot/grub` using any of the command line text editor such as vi. Look for the line 'splashimage=' change the path of splash screen image from `/grub/splash.xpm.gz` to `/grub/photo.xpm.gz` (as in our example).

Reboot your system to see the change in GRUB's screen. You can avoid the task of converting image to xpm format by using readily available splash-screen images from the Net. Just save the image in `/boot/grub` and change the path of splash image in the `grub.conf` file. Flaunt the new look!

PowerToys For Windows XP

The Windows XP PowerToys is a set of applications you can download for free from Microsoft. They are not officially part of Windows XP, but Microsoft makes them available as enhancements to it. One could say that the functionality the PowerToys provide is something that should have been available in Windows all along, but for some reason is not. As such, several of the PowerToys could be considered essential to the OS.

The XP PowerToys site is www.microsoft.com/windowxsp/downloads/powertoys/xp/powertoys.mspx

The following is a list of the PowerToys you can download from there.

Color Control Panel Applet For Windows XP

Professional-level photographers and designers know that getting consistent, accurate colour from file to screen to print and beyond is a requirement for great results. On XP, the means to obtain great colour is built into the OS via Image Color Management (ICM) 2.0. ICM 2.0 is an International Color Consortium (ICC)-

